

Set 9 - Classes and Objects (Basics)

Q1.

Create a Java class named Product with two attributes: productName (String) and price (double). Create an object of Product in your main method and print its details.

Answer:

```
public class Product {
    String productName;
    double price;

    public static void main(String[] args) {
        Product product = new Product();
        product.productName = "Bluetooth Speaker";
        product.price = 1499.0;

        System.out.println("Product Name: " + product.productName);
        System.out.println("Price: Rs. " + product.price);
    }
}

/* Sample Output:
Product Name: Bluetooth Speaker
Price: Rs. 1499.0
*/
```

Q2.

Build a Java class called Playlist that represents a Spotify playlist with attributes: playlistName (String) and totalSongs (int). Add a method displayInfo() that prints both attributes in a user-friendly format.

Answer:

```
public class Playlist {
    String playlistName;
    int totalSongs;

    void displayInfo() {
        System.out.println("Playlist: " + playlistName + " | Total Songs: " + totalSongs);
    }

    public static void main(String[] args) {
        Playlist playlist = new Playlist();
        playlist.playlistName = "Chill Vibes";
        playlist.totalSongs = 32;
        playlist.displayInfo();
    }
}

/* Sample Output:
Playlist: Chill Vibes | Total Songs: 32
*/
```

Q3.

Imagine you are designing a Flipkart shopping cart. Create a CartItem class with attributes: itemName (String) and quantity (int). Write a method increaseQuantity() that adds 1 to the quantity and prints the updated value. Hint: Call increaseQuantity() twice from main and observe the output.

Answer:

```
public class CartItem {
    String itemName;
    int quantity;

    void increaseQuantity() {
        quantity++;
        System.out.println(itemName + " quantity updated to: " + quantity);
    }

    public static void main(String[] args) {
        CartItem item = new CartItem();
        item.itemName = "Wireless Charger";
        item.quantity = 1;

        item.increaseQuantity();
        item.increaseQuantity();
    }
}

/* Sample Output:
Wireless Charger quantity updated to: 2
Wireless Charger quantity updated to: 3
*/
```

Q4.

Create a Java class called MovieTicket for a BookMyShow-style app, with attributes: movieTitle (String) and seatNumber (String). Add a constructor to initialize both attributes and a method printTicket() to display the ticket details.

Answer:

```
public class MovieTicket {
    String movieTitle;
    String seatNumber;

    public MovieTicket(String movieTitle, String seatNumber) {
        this.movieTitle = movieTitle;
        this.seatNumber = seatNumber;
    }

    void printTicket() {
        System.out.println("---- Movie Ticket ----");
        System.out.println("Movie: " + movieTitle);
        System.out.println("Seat: " + seatNumber);
    }

    public static void main(String[] args) {
        MovieTicket ticket = new MovieTicket("Avengers: Endgame", "G12");
        ticket.printTicket();
    }
}

/* Sample Output:
---- Movie Ticket ----
Movie: Avengers: Endgame
```

Seat: G12

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